



2016 中国互联网络安全大会
China Internet Security Conference

协同联动 共建安全+命运共同体

How to build resilience to cyberattacks

Jonathan Reiber

Senior Fellow
UC Berkeley Center for
Long-term Cybersecurity



Jonathan,
Compelling me to
seize an opportunity
and to change places
with! Raymond

Jonathan -
I've one more!
In the time of a year
I'll be a son of mine!
I'll be a son of mine!
I'll be a son of mine!
I'll be a son of mine!

Jonathan -
I'll be a son of mine!
I'll be a son of mine!
I'll be a son of mine!
I'll be a son of mine!

Jonathan,
Thanks for all your
great work in D.O.
You really made a
difference and I am
going to miss my longtime
school buddy. Looking
forward to seeing you
do great things in the
West Coast!
- Andrew

Ramon - stay cool
Christie

Jonathan,
It has been a pleasure
and an honor to work
with you. You are an
incredibly talented
person with the right
perspective at the right
time, especially at this
critical time.
Very best to you my friend.
Fair winds following fear
Ed Devaney

Jonathan - Thanks for
being a great colleague
and friend and introducing
me to the world of aces and
pens. Best luck on your
next and many adventures!
- Kaitlyn

Jonathan,
Thanks for all of
your hard work and
dedication to our
national security.
I'll be a son of mine!
I'll be a son of mine!
I'll be a son of mine!

Jonathan,
Thank you for
your tremendous
contribution to our
national security.
I'll be a son of mine!
I'll be a son of mine!
I'll be a son of mine!

Jonathan,
Thank you for
your service (and
glad we are
holding on to
you in some
small way!).
Best,
Zach

Jonathan,
All thanks
to you for your
service (and
glad we are
holding on to
you in some
small way!).
Best,
Zach

Jonathan,
I'll be a son of mine!
I'll be a son of mine!
I'll be a son of mine!



A. Nelson Davis

Jonathan,
Thanks for all of
your hard work and dedication
to the cyber mission and D.O.
Your contributions are numerous
and passion infectious.
All the best
- Kaitlyn

Jonathan,
I'm going to miss
you my friend.
Your dedication to this
place was unmatched.
I'll see you in
the best way!
- Kaitlyn

Jonathan -
Thank you for
being a great
colleague's friend.
The many clubs will
miss your laughter and
presence. Please keep
in touch - I'll be there!
- Kaitlyn

Jonathan,
I'll be a son of mine!
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Jonathan,
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I'll be a son of mine!
I'll be a son of mine!

Jonathan,
Congratulations on this
awesome opportunity! The boy
area is an awesome place to
live! Raymond

R. J. [unclear] -
 You have to get a [unclear]
 for the [unclear] [unclear] [unclear]
 The [unclear] [unclear] [unclear]
 [unclear] [unclear] [unclear] [unclear]
 [unclear] [unclear] [unclear] [unclear]
 [unclear] [unclear] [unclear] [unclear]

Jonathan -
 Stoked for this
 opportunity for you &
 your proximity to
 great waves. Think of
 us when you're surfing
 Cap! Love
 D + S

Jonathan,
Thanks for all your
great work in Dot.
You really made a
difference and I am
going to miss my longtime
School buddy. Looking
forward to seeing you
do great things on the
West Coast!

Reiben - stay cool!
Christel

Jonathan

Jonathan:
It has been a pleasure
and an honor to work
with you. You are an
incredibly talented
person with the right
perspective at the right
time, especially as I
pertain to family. All the
very best to you my friend.
Fair Winds Following Sea
Ed Devaney

Jonathan - Thanks for
being a great colleague
and friend and introducing
me to the world of ones and
zeros. Best luck on your
next and many adventures!
- Krispa

Jonathan
Thanks for all of
your hard work outside
these years. Congrat
on the new opportunity
I hope you enjoy your new
All the best - Good luck,
W. B. ~~W. B.~~ ^{W. B.}
in your
best of luck

you were a tremendous
front office colleague and
a great friend. Thanks to all
the people who worked with you.
- Bob

Jonathan,
Thank you for
your tremendous
contributions to our
national security -
cyber and much else!
It's been a great
pleasure and an honor
to work with you!
Warmest regards, Jim

Josephson,

Thanks for
your service (and
glad we are
holding on to
you in some
small way!)

Jonathan,

It has been
a pleasure knowing
you Good luck in
your next adventure

Tomato for

Thanks
for support
of our future
ambitions
May
Cont

May the collective
continue to thrive!
Best, Paul S.

Thanks
 I've been
 It's been a
 also with you
 writing with you
 best of luck to
 you & the
 to make time
 the all these
 Cheers
 [Signature]



A. Nelson Davis

Touchan,
I envy you the new
like you're about to
undertake! You did well
for yourself, says! Give
back to Kari.

Jonathan -
Great lady
on the left
cuss!
Rebo I.

Johnston: My been a great
joy for me to have the chance
work with you on the Cofesa
in FE. Thanks for a happy
collab. Hope to cross paths
in the future.
- Don
Mell

Pick over
 looking out
 again soon. - Chris

- Great working with you
 we will do the work
 - "Good job!" But
 - "Good job!" But
 - "Good job!" But

John
I don't
know
how to stay
here
I still need to get that
- Matt Myers

about
leaves
miss
friend.
Hila

corollas.
History with
engaged working with
and hope to have the
operation of group. Ben
close!
Mike Clark

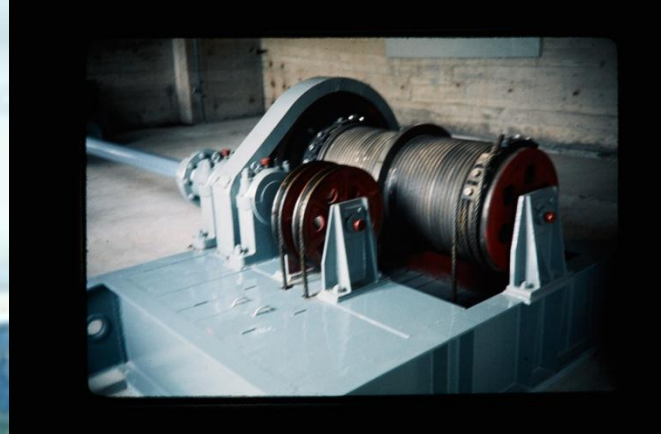
to send with you, and a greater one to
L. There are also some other things.

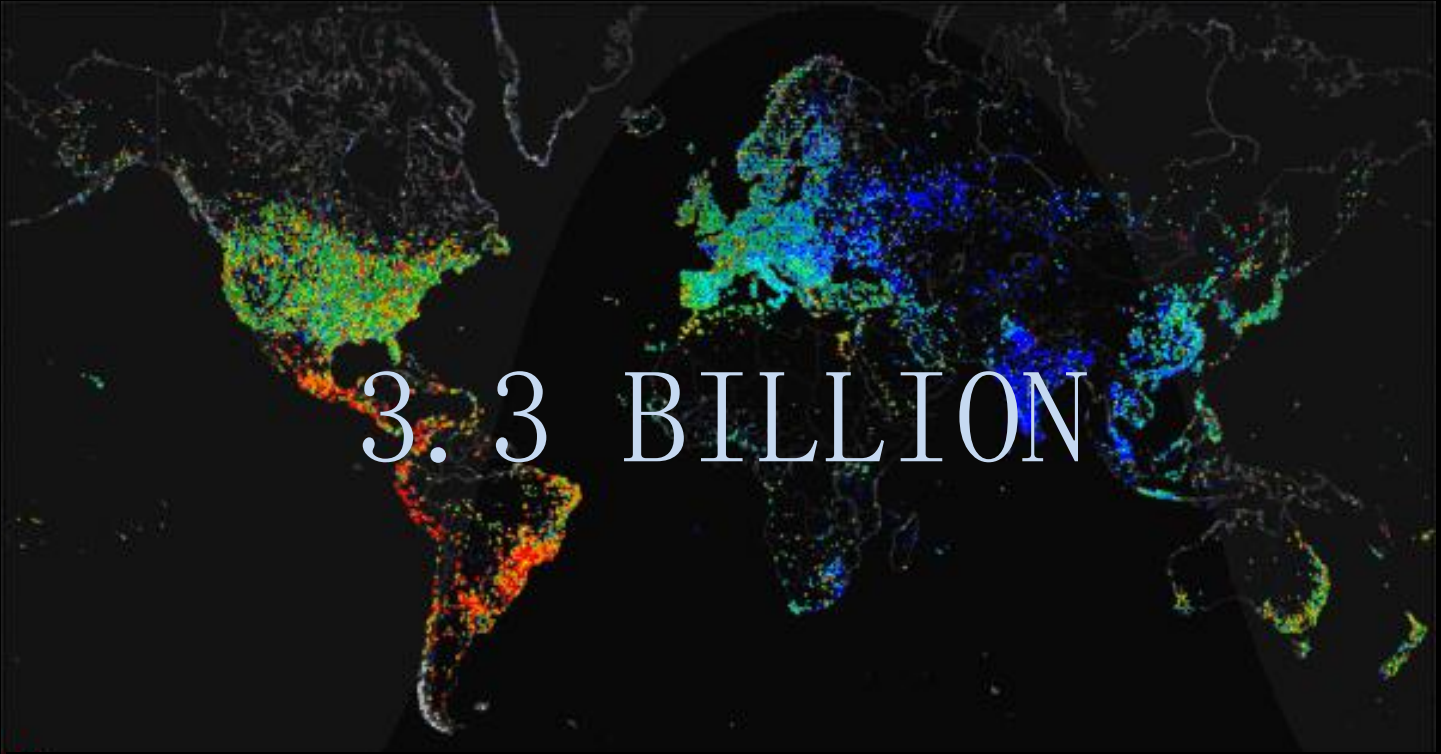
A. Nelson Davis



High consequence risks





A world map with a dark background, showing population density. The landmasses are outlined in a light gray. The map is covered with a dense layer of small, multi-colored dots (red, orange, yellow, green, blue) representing population density. The dots are most concentrated in North America, Europe, and East Asia. The text "3.3 BILLION" is overlaid in the center of the map.

3.3 BILLION

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4. 谢永江-网络漏洞的法律属性



Sony Pictures Entertainment...

Hacked By #GOP

Warning :

We've already warned you, and this is just a beginning.

We continue till our request be met.

We've obtained all your internal data including your secrets and top secret.

If you don't obey us, we'll release data shown below to the world.

Determine what will you do till November the **24th, 11:00 PM(GMT)**.

Data Link :

<https://www.sonypicturesstockfootage.com/SPEData.zip>

<http://dmipiaewh36.spe.sony.com/SPEData.zip>

<http://www.ntcnt.ru/SPEData.zip>

<http://www.thammasatpress.com/SPEData.zip>

<http://moodle.universidadebematech.com.br/SPEData.zip>

We Are Anonymous.

We Are Legion.

We Do Not Forgive.

We Do Not Forget.

Expect Us.



A crossroads of two futures....

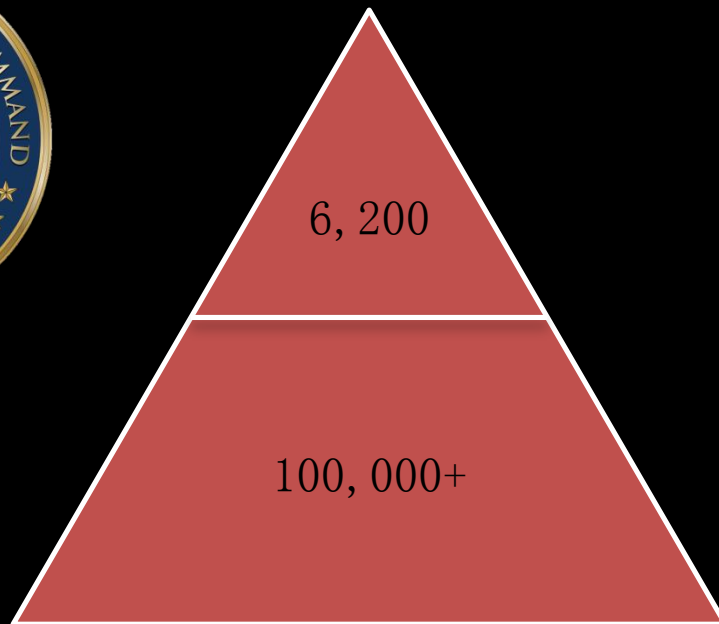
Pervasive destructive attacks...

A crossroads of two futures....

Pervasive destructive attacks... *or*

...cultures of cybersecurity and
resilience that take root

Stakeholders





Wed, Apr 27

🔒 Messages you send to this chat and calls are now secured with end-to-end encryption. Tap for more info.

Today



10:40 AM ✓✓

(Live long and prosper via encryption, good people.)

10:41 AM ✓✓

Technology is not the center of the story...

...that is us

Four recommendations

- Every organization needs a cyberstrategy

Four recommendations

- Every organization needs a cyberstrategy
- Get the basics right

Four recommendations

- Every organization needs a cyberstrategy
- Get the basics right
- Government and private sector *together*

Information sharing; penetration testing; future of partnerships

1. *The Sharing of Cyber Threat Indicators and Defensive Measures by the Federal Government Under the Cybersecurity Information Sharing Act of 2015*
 - Explains how government shares info about cyber threat indicators (CTIs) and defensive measures (DMs)
 - Some organizations have security clearances; some don't
 - Federal government declassifies information often
 - Agencies share information about threats and best practices
2. *Guidance to Assist Non-Federal Entities to Share Cyber Threat Indicators and Defensive Measures with Federal Entities under the Cybersecurity Information Sharing Act*
 - Helps non-federal government institutions determine DMs, and explains liability and statutory protections
 - Explains nature of cyber threat indicators and defensive measures
 - Protected health, human resources, financial, consumer information
 - Options for non-federal entities to report: Automated Indicator Sharing (AIS) system, DHS web form, and Information Sharing and Analysis Organizations and Centers (ISAOs and ISACs)

3. *The Final Procedures Related to the Receipt of Cyber Threat Indicators and Defensive Measures by the Federal Government*

- Methods for information sharing
- Human reviewers analyze the information
- Covers sanctions that arise from violations of the requirements, which include “remedial training; loss of access to information; loss of a security clearance; and termination of employment”

4. *The Privacy and Civil Liberties Final Guidelines: Cybersecurity Information Sharing Act of 2015*

- Protecting privacy and civil liberties in CISA-compliant sharing of CTIs and DMs
- Reiterates the Fair Information Practice Principles
- Notes that DMs should include no info that can identify specific individuals
- Guidance on what info notices that deal with non-cyber threat indicators should contain, and the circumstances under which information about CTIs and DMs should be retained

How do Internet and private companies enterprises share cyber threat indicators and defensive measures with government?

- Ad hoc : between companies and govies that trust each other
- Automated Indicator Sharing (AIS), through DHS
 - Standardized fields for input of relevant info
 - Info only shares once analyzed
 - DHS also provides an online form that businesses can fill out
- Email to National Cybersecurity and Communications Integration Center (NCCIC) with information about CTIs and DMs.
- Information Sharing and Analysis Centers (ISACs) and Information Sharing and Analysis Organizations (ISAOs) who communicate with the federal government on their behalf

- ISAOs and ISACs
 - ISACs: funded and operated by private enterprises
 - Connected to critical infrastructure sectors (IT ISAC, health ISAC)
 - ISAOs: share info among small business, consulting, accounting, legal firms
 - Internet companies: encouraged to join ISAOs through workshops and public meetings that are held around the country.
- Public forums: experts share information
 - White House Cybersecurity Commission; Congress
- Cyber information Sharing and Collaboration Program
 - Companies gain access to NCCIC floor at DHS; classified clearances
- DHS's [Enhanced Cybersecurity Services](#) (ECS) shares info with commercial ISPs; open to all public and private entities in the United States.
- Info and collaboration with US-CERT
- Enduring Security Framework
- Ad hoc information sharing and collaboration: vital! Trust!

Penetration testing

- Penetration testing roles
 - *Also known as “red-teaming” or “tiger teaming”*
 - Purpose: emulate & break in to nets to find vulnerabilities
- Key difference between attacker/pen-tester: **permission**
 - Have auditor protected by legal counsel
- Contracts should protect; be specific
- Diverse ways of managing pen testers:
 - Internal; external firms; bug bounties
- Governance: Certified Penetration Tester (CPT).
 - Certifications: CISSP; CEH.
- Wassenaar Agreements: legal regimes could impose limits on on dual-use export controls to do basic monitoring
- Need to be careful about how we control pen testing



Senior Security Analyst - Penetration

11d

Mastercard

New York City, NY



Advisory Senior Consultant –

18d

EY

US



Paranoids Penetration Tester

15d

Yahoo

New York, NY , USA



Sr. Information Security Engineer -

3d

LinkedIn

Mountain View

 Apply

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USA: 1-800-708-2164

+61 1300 031337 (International)

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Four recommendations

- Every organization needs a cyberstrategy
- Get the basics right
- Government and private sector *together*
- Plan for resilience

And then...

Put down your phones.

And then...

Put down your phones.

Our social bonds *cannot be hacked.*





THANKS



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360互联网安全中心